

Adrenal Insufficiency/CAH — Chronic AI Clinic & Adrenal Crisis ER

SKIN COLOR CLUE

PRIMARY AI
POMC HIGH

SECONDARY AI
POMC LOW

PRIMARY AI
POMC HIGH

NO PIGMENT
NO HYPERK

CHRONIC AI CLINIC

FATIGUE WEAKNESS

FATIGUE WEAKNESS

ACTH

MSH

POMC DERIVED

BUCCAL MUCOSA

POSTURAL HYPOTENSION

SALT CRAVING

ABDOMINAL PAIN
NAUSEA

NA 126 LOW

K 5.9 HIGH

GLUCOSE 55 LOW

MILD METABOLIC ACIDOSIS

BUN AND CR UP
VOLUME DEPLETION

CBC

ADRENAL ANDROGEN DEFICIENCY

DRY ITCHY SKIN

LOW LIBIDO

LOSS OF AXILLARY AND PUBIC HAIR

ADRENAL ANDROGEN DEFICIENCY

HYPERPIGMENTED PALMAR CREASES

POSTURAL HYPOTENSION

KNUCKLES

SCAR PIGMENTATION

MYALGIA JOINT PAIN

WEIGHT LOSS ANOREXIA

LOW-GRADE FEVER

TIRED THIN TAN
HYPOTENSIVE
HYPERKALEMIC

WOMEN'S ANDROGEN DEFICIENCY

DRY ITCHY SKIN

LOW LIBIDO

LOSS OF AXILLARY AND PUBIC HAIR

ADRENAL ANDROGEN DEFICIENCY

ADRENAL CRISIS ER BAY

CRISIS TRIGGERS

INFECTION

SURGERY OR TRAUMA

ABRUPT STEROID STOP

PITUITARY OR ADRENAL SURGERY

BILATERAL ADRENAL HEMORRHAGE

WATERHOUSE-FRIDERICHSEN VIGNETTE

MENINGOCOCCEMIA

PURPURA FULMINANS

BILATERAL ADRENAL HEMORRHAGE

HYDROCORTISONE + FLUIDS + ANTIBIOTICS

IMMEDIATE TRIPLE THERAPY

EMERGENCY TREATMENT CART

DO NOT WAIT FOR LABS

IV HYDROCORTISONE

IV NORMAL SALINE

DEXTRASE

IF HYPOGLYCEMIC

TREAT TRIGGER

STEP 1

STEP 2

STEP 3

STEP 4

CRISIS LAB PANEL

NA DOWN

K UP IF PRIMARY

GLUCOSE DOWN

CORTISOL DOWN

ACTH UP IF PRIMARY

TREATMENT FIRST LABS SECOND

CHRONIC AI CAN DECOMPENSATE AT ANY TIME

CONFUSION

HYPOGLYCEMIA

FEVER

ABDOMINAL PAIN

SHOCK

N AND V

CHRONIC AI EQUALS TIRED THIN TAN HYPOTENSIVE HYPERKALEMIC. CRISIS EQUALS SHOCK PLUS HYONATREMIA PLUS HYPERKALEMIA PLUS HYPOGLYCEMIA. GIVE HYDROCORTISONE AND SALINE NOW DO NOT WAIT.

1. Skin color clue

WHAT YOU SEE

Top-left SKIN COLOR CLUE: primary hyperpigmented vs secondary pale

WHAT IT ENCODES

PRIMARY AI (Addison, adrenal destroyed): cortisol falls → loss of negative feedback → pituitary ramps up POMC, which is cleaved into BOTH ACTH and alpha-MSH; MSH stimulates melanocytes → diffuse hyperpigmentation. SECONDARY AI (pituitary/hypothalamic, low ACTH): no excess MSH → pale skin, and aldosterone is preserved (RAAS-driven) so NO hyperkalemia and minimal salt-wasting.

BOARD TRAP

Hyperpigmentation = primary AI only (high ACTH/MSH). A pale patient with low cortisol AND low/normal ACTH = secondary, and they will NOT be hyperkalemic.

2. Chronic AI clinic

WHAT YOU SEE

CHRONIC AI CLINIC heading over the tired patient

WHAT IT ENCODES

Chronic primary AI is an insidious, nonspecific picture: tired, thin, tan, hypotensive, hyperkalemic. Etiologies — in the developed world #1 is AUTOIMMUNE adrenalitis (21-hydroxylase autoantibodies, often part of APS-1/APS-2); WORLDWIDE the leading cause is TB. Other causes: metastases, adrenoleukodystrophy, drugs (ketoconazole, etomidate).

BOARD TRAP

Autoimmune is the #1 cause in the West; TB is #1 worldwide — board picks the answer to fit the geographic/clinical clue.

3. Fatigue & weakness

WHAT YOU SEE

Labels fatigue, weakness, low-grade fever around the patient

WHAT IT ENCODES

Nonspecific chronic symptoms: fatigue, weakness, weight loss/anorexia, myalgia/arthralgia, low-grade fever. Because these are vague, Addison is frequently missed until a stressor tips the patient into crisis.

4. Buccal mucosa pigment

WHAT YOU SEE

Buccal mucosa + knuckle hyperpigmentation labeled, ACTH/MSH

WHAT IT ENCODES

ACTH is co-secreted with alpha-MSH (both from POMC); chronically high ACTH/MSH pigments sun-exposed skin, BUCCAL/gingival mucosa, knuckles/extensor surfaces, palmar creases, scars, and the areolae. Mucosal and palmar-crease pigment is the most specific sign because it is not explained by sun exposure.

BOARD TRAP

Hyperpigmentation of buccal mucosa/palmar creases is the classic exam clue for chronic primary (not secondary) AI.

5. Postural hypotension

WHAT YOU SEE

Postural hypotension + salt craving + nausea labels

WHAT IT ENCODES

Aldosterone deficiency → renal Na⁺ and water loss → volume depletion → orthostatic hypotension and SALT CRAVING; cortisol loss adds GI symptoms (nausea, vomiting, anorexia, abdominal pain) and impaired vascular tone/pressor responsiveness.

6. Electrolyte panel

WHAT YOU SEE

Lab box: low Na, high K, low glucose, metabolic acidosis

WHAT IT ENCODES

Chronic primary AI labs: HYPOnatremia (cortisol loss → impaired free-water excretion/SIADH-like + aldo loss), HYPERkalemia + non-anion-gap metabolic acidosis (aldosterone loss), and HYPOglycemia (cortisol is counter-regulatory). May see mild hypercalcemia and eosinophilia.

BOARD TRAP

Hyperkalemia points to PRIMARY AI (aldosterone loss). Secondary AI keeps aldosterone (RAAS-driven) so K⁺ is normal — hyperkalemia rules secondary OUT.

7. Women's androgen deficiency

WHAT YOU SEE

Women's androgen deficiency panel: low libido, loss of axillary/pubis hair, dry skin

WHAT IT ENCODES

The adrenal is the main source of androgens (DHEA/DHEA-S) in women; in primary AI women lose libido, develop loss of axillary/pubis hair and dry skin. Men are unaffected here because testes supply testosterone.

8. Crisis = shock + electrolytes

WHAT YOU SEE

ADRENAL CRISIS ER BAY heading, red alarm

WHAT IT ENCODES

Adrenal (Addisonian) crisis = the lethal acute form: distributive/hypovolemic SHOCK + hyponatremia + hyperkalemia + hypoglycemia, often with fever and abdominal pain mimicking an acute abdomen. It is an endocrine emergency.

9. Crisis triggers

WHAT YOU SEE

CRISIS TRIGGERS: infection, surgery/trauma, abrupt steroid stop, pituitary apoplexy, bilateral adrenal hemorrhage

WHAT IT ENCODES

Precipitants: infection/sepsis, surgery/trauma/major stress, ABRUPT withdrawal of chronic exogenous steroids (suppressed HPA axis can't respond), pituitary apoplexy, and bilateral adrenal hemorrhage. The classic increased cortisol demand of any acute illness in an AI patient unmasks the deficiency.

BOARD TRAP

Abruptly stopping long-term glucocorticoids (or failing to stress-dose during illness/surgery) is the single most testable precipitant of crisis.

10. Waterhouse-Friderichsen

WHAT YOU SEE

Top-right WATERHOUSE-FRIDERICHSEN vignette: meningococcemia/purpura fulminans → bilateral adrenal hemorrhage

WHAT IT ENCODES

Waterhouse-Friderichsen syndrome = fulminant bilateral adrenal HEMORRHAGE causing acute primary AI, classically from *Neisseria meningitidis* sepsis (also *Pseudomonas*, *pneumococcus*, *Haemophilus*). Presents with shock, DIC and a petechial/purpuric (purpura fulminans) rash — treat sepsis AND give stress-dose steroids.

BOARD TRAP

A child/young adult with meningococcemia, purpura and refractory shock = Waterhouse-Friderichsen; the hypotension won't resolve until you give hydrocortisone.

11. Hypotension / shock

WHAT YOU SEE

Collapsed patient on stretcher labeled shock, abdominal pain, fever, confusion

WHAT IT ENCODES

Hallmark = refractory HYPOTENSION/shock that responds poorly to fluids and vasopressors alone (cortisol is permissive for catecholamine vascular tone), plus abdominal pain, fever and confusion. Pressor-resistant shock should prompt empiric hydrocortisone.

BOARD TRAP

Vasopressor-refractory shock + suggestive history = give steroids; don't keep escalating pressors.

12. DO NOT WAIT FOR LABS

WHAT YOU SEE

Emergency treatment cart: DO NOT WAIT FOR LABS

WHAT IT ENCODES

Treatment is empiric and IMMEDIATE — draw a cortisol/ACTH if convenient but do NOT delay therapy for results. Untreated crisis is rapidly fatal; the consequences of treating an equivocal case are negligible.

BOARD TRAP

Waiting for the cortisol/ACTH result (or for a cosyntropin stim test) before treating a suspected crisis is the cardinal board trap — treat first, confirm later.

13. IV hydrocortisone

WHAT YOU SEE

IV HYDROCORTISONE syringe on the cart

WHAT IT ENCODES

Give IV HYDROCORTISONE 100 mg bolus immediately, then 50 mg q6h (or 200 mg/24h infusion). Hydrocortisone is preferred because at stress doses it has enough mineralocorticoid activity to cover aldosterone too.

BOARD TRAP

If you must preserve a pending cortisol/cosyntropin test, give DEXAMETHASONE instead — it does not cross-react in the cortisol assay; hydrocortisone would falsely elevate it.

14. IV saline + dextrose

WHAT YOU SEE

IV normal saline and dextrose bags labeled

WHAT IT ENCODES

Aggressive IV isotonic NORMAL SALINE (often 1–3 L rapidly) corrects volume depletion and hyponatremia; add DEXTROSE for hypoglycemia. Hyperkalemia usually corrects with volume + cortisol/aldosterone effect and rarely needs specific treatment.

BOARD TRAP

Correct the volume/Na with normal saline — do not over-rapidly correct hyponatremia, and the hyperkalemia generally self-corrects once steroids/fluids are on board.

15. Treat the trigger

WHAT YOU SEE

Bottom note: treat trigger; crisis lab panel for Na/K/glucose/cortisol

WHAT IT ENCODES

After resuscitation, identify and treat the precipitant (e.g., antibiotics for the infection). High-dose hydrocortisone supplies the mineralocorticoid need acutely; once tapered below ~50 mg/day, add fludrocortisone for ongoing replacement in primary AI.

BOARD TRAP

At stress-dose hydrocortisone you do NOT need separate fludrocortisone acutely — its intrinsic mineralocorticoid effect is sufficient; add fludrocortisone only as the dose is tapered.